of FMEA.
1.6 Whilst FMEA is suggested as one of the most flexible analysis techniques, it is accepted that there are other methods which may be used and which in certain circumstances may offer an equally comprehensive insight into particular failure characteristics.
2 QQiectives
2.1 The primary objective of FMEA is to provide a comprehensive, systematic and documented investigation which establishes the important failure
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conditions of the craft and assesses their significance with regard to the safety of the craft, its occupants and the environment.
2.2 The main aims of undertaking the analysis are to:

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conditions of the craft and assesses their significance with regard to the safety of the craft, its occupants and the environment.
2.2 The main aims of undertaking the analysis are to:
.1 provide the Administration with the results of a study into the craft's failure characteristics so as to assist in an assessment of the levels of safety proposed for the craft's operation;
.2 provide craft operators with data to generate comprehensive training, operational and maintenance programmes and documentation; and
.3 provide craft and system designers with data to audit their proposed designs.
3 S c o p e of app li c a t i on
3.1 FMEA should be conducted for each high speed craft, before its entry into service, in respect of the systems as required under the provisions of 5.2, 9.1.10, 12.1.1 and 16.2.6 of this Code.
3.2 For craft of the same design and having the same equipment, one FMEA on the lead craft will be sufficient, but each of the craft should be subject to the same FMEA conclusion trials.
4 System failure mode and effects anal~

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3.2 For craft of the same design and having the same equipment, one FMEA on the lead craft will be sufficient, but each of the craft should be subject to the same FMEA conclusion trials.
4 System failure mode and effects anal~
4.1 Before proceeding with a detailed FMEA into the effects of the failure of the system elements on the system functional output it is necessary to perform

a functional failure analysis of the craft's important systems. In this way only systems which fail the functional failure analysis need to be investigated by a more detailed FMEA.

4.2 When conducting a system FMEA the following typical operational modes within the normal design environmental conditions of the craft should be considered:

.1 normal seagoing conditions at full speed;

.2 maximum permitted operating speed in congested waters; and

.3 manoeuvring alongside.

4.3 The functional interdependence of these systems should also be described in either block diagrams or fault tree diagrams or in a narrative format to enable the failure effects to be understood. As far as applicable, each of the systems to be analysed is assumed to fail in the following failure modes:

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in either block diagrams or fault tree diagrams or in a narrative format to enable the failure effects to be understood. As far as applicable, each of the systems to be analysed is assumed to fail in the following failure modes:

.1 complete loss of function;

.2 rapid change to maximum or minimum output;

.3 uncontrolled or varying output;

 .4 premature operation;

.5 failure to operate at a prescribed time; and

.6 failure to cease operation at a prescribed time.

Depending on the system under consideration other failure modes may have to be taken into account.

4.4 If a system can fail without any hazardous or catastrophic effect, there is no need to conduct a detailed FMEA into the system architecture. For systems whose individual failure can cause hazardous or catastrophic effects and where a redundant system is not provided, a detailed FMEA as described in the following paragraphs should be followed. Results of the system functional

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systems whose individual failure can cause hazardous or catastrophic effects and where a redundant system is not provided, a detailed FMEA as described in the following paragraphs should be followed. Results of the system functional failure analysis should be documented and confirmed by a practical test programme drawn up from the analysis.

4.5 Where a system, the failure of which may cause a hazardous or catastrophic effect, is provided with a redundant system, a detailed FMEA may not be required provided that:

.1 the redundant system can be put into operation or can take over the failed system within the time-limit dictated by the most onerous

operational mode in 4.2 without hazarding the craft;
.2 the redundant system is completely independent from the system and does not share any common system element the failure of which would cause failure of both the system and the redundant system. Common system element may be acceptable if the probability of failure complies with section 13; and
.3 the redundant system may share the same power source as the system. In such case an alternative power source should be readily available

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system element may be acceptable if the probability of failure complies with section 13; and
.3 the redundant system may share the same power source as the system. In such case an alternative power source should be readily available with regard to the requirement of .1.
The probability and effects of operator error to bring in the redundant system should also be considered.
5 Equipment failure mode and effects analysis
The systems to be subject to a more detailed FMEA investigation at this stage should include all those that have failed the system FMEA and may include those that have a very important influence on the safety of the craft and its occupants and which require an investigation at a deeper level than that undertaken in the system functional failure analysis. These systems are often those which have been specifically designed or adapted for the craft, such as the craft's electrical and hydraulic systems.
6 Procedures
The following steps are necessary to perform FMEA:
.1 to define the system to be analysed;
.2 to illustrate the interrelationships of functional elements of the system by means of block diagrams;
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.3 to identify all potential failure modes and their causes;
.4 to evaluate the effects on the system of each failure mode:
.5 to identify failure detection methods:
.6 to identify corrective measures for failure modes:
.7 to assess the probability of failures causing hazardous or

catastrophic effects, where applicable:
.8 to document the analysis:
.9 to develop a test programme:
.10 to prepare FMEA report.
7 System definition
The first step in an FMEA study is a detailed study of the system to be analysed through the use of drawings and equipment manuals. A narrative description of the system and its functional requirements should be drawn up including the following information:
.1 general description of system operation and structure;
.2 functional relationship among the system elements:

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description of the system and its functional requirements should be drawn up including the following information:
.1 general description of system operation and structure;
.2 functional relationship among the system elements:
.3 acceptable functional performance limits of the system and its constituent elements in each of the typical operational modes: and
.4 system constraints.
8 ~velopment of system block diagrruns